

Rowan McAlpin

[ADDRESS REDACTED]

ROWAN@MCALPIN.DEV

<https://github.com/rowan-mcalpin>

Skills

- Project management
- CAD
- Mechanical & Electrical Manufacturing
- Computer programming including Java, Kotlin, etc.
- Version control systems (Git/GitHub)

Experience

Cambria Corporation (2026)

Junior Engineer (2026-Present, currently not working due to college; will resume in summer)

- Work with senior engineer to create PCB layouts (using Altium)
- Assemble, configure, and program industrial manufacturing robots
- Sensor manufacturing including hand soldering, large SMT solder machines, ultrasonic welders, and automatic point soldering machines
- Facilitate rigorous sensor testing and quality control
- Work with senior staff & company executives to improve workflows and efficiency

NextFTC Software Library (see: <https://nextftc.dev>) (2022-Present)

Lead Developer (2022-Present)

- Oversee the contributions of other developers
- Rapidly respond to issues in the code
- Assist teams with debugging issues and using the library
- Moderate a Discord support server with 450 users

Atomic Robotics *FIRST* Tech Challenge team (2021-2026)

Team Captain (2024-2026)

- Lead & manage a 16-person team consisting of middle and high schoolers
- Oversee & lead the CAD, manufacturing, and programming of a complex competitive robot
- Maintain the software library I developed (NextFTC) which is used by ~100 teams across the globe in 10+ countries

Software Lead (2022-2026)

- Lead software development of a complex robot, using Kotlin and Java
- Develop & maintain a software library

Software Team (2021-2022)

- Learn and develop Kotlin and Java code to control a complex robot

Chill Out *FIRST* Robotics Competition team (2025-2026)

Vice President (2026)

- Manage a 64-member high school robotics team
- Oversee the design, fabrication, and programming of a 135-pound complex machine
- Operate industrial manufacturing equipment including CNC, lathe, etc.

Software & Manufacturing (2025)

- Assist in the manufacturing & programming of a complex robot
- Ended season ranked 16th worldwide out of nearly 4,000 teams

West Coast Invitational robotics competition (see: <https://westcoastinvitational.org/>)

(2024-Present)

Founder, Event Manager, Executive (2024-Present)

- Founded an international offseason robotics competition for the *FIRST* Tech Challenge
- Lead the planning and execution of the event including volunteer and team coordination, A/V, venue interfacing, and financing

Education

North Seattle College - FALL 2024-SPRING 2026

Cumulative GPA (unweighted): 3.980

Clubs/activities: committee member of the North Seattle College Computer Science Club (2024-2026)

Edmonds Heights K-12 - FALL 2022-SPRING 2026

Cumulative GPA (unweighted): 4.0

Awards

FIRST Tech Challenge Leadership Award Semi-Finalist (2024)

An award given by FIRST Robotics to recognize outstanding student leaders within the FIRST Tech Challenge (FTC) program

North Seattle College President's List (2025, 2026)